Parsing PSD documents

- Available Options
  - Official APIs
  - Top npm modules
- AG-PSD library POC
  - What can we extract?
    - Artboards
    - Groups
    - Layers
    - Text support
  - Limitations
    - Text/Fonts
    - Components configurations
    - Layer's border on particular side
  - POC and results
- Official APIs POC
  - Results PSD to VDX Studio
  - What can we extract by raw "get" command?
    - Artboard JSON
    - Group JSON
    - Layer JSON
    - Text Layer JSON
    - Rectangle Layer JSON
- Conclusion

Available Options

Official APIs

- https://developer.adobe.com/photoshop/
- https://developer.adobe.com/photoshop/photoshop-api-docs/api/#tag/Photoshop/operation/documentManifestStatus

Top npm modules

PSD: https://github.com/meltingice/psd.js#readme

AG-PSD: https://github.com/Agamnentzar/ag-psd/blob/master/README_PSD.md

AG-PSD is preferable as it provides more detailed information about text layers, effects, etc.

> Can be used on both client and server sides

AG-PSD library POC

What can we extract?

Artboards

Useful information:

- Name
- Frame rectangle: top, left, right, bottom in absolute pixels
- Content(child groups/layers)

We can use artboards to define product/Ad size design

Groups

Useful information:

- Name
- Content(child groups/layers)

We can use groups to define design parts of Ad unit, e.g. header section, tabs, components, slides, etc.

There is no frame rectangle information provided by ag-psd, but we can calculate all of those from group's content layers(PSD.js lib and Official APIs provide group frame details).

Layers

Useful information:

- Name
- Frame rectangle: top, left, right, bottom in absolute pixels
- Image data
- Effects(bevel, stroke, glows, color overlay, etc.)

We can use layer's image data for backgrounds, logos and buttons.

Shapes are supported as well, we can use rectangle to define borders and background fill color.

Text support

Useful information:

- Transform matrix (2d)

- Font family
- Font size
- Font color

- Text alignment
- Text leading (line height)
- Text tracking (letter spacing)
- Text vertical/horizontal scale
- Text baseline shift

- Text superscript
- Text subscript
- Text underline
- Text strikethrough

Styling of text parts supported as well.

Limitations

Text/Fonts

Fonts are not a part of psd document, it will be required to validate/upload used fonts.

Font names may not match, e.g. in Photoshop "Arial", while in exported JSON "ArialMT". We need exact font name to compose font-face in css.

Font styles regular/bold/italic/bold italic will require separate font files. Faux bold/italic or “synthesized” font styles can’t be used, as there is no way to apply the same by css with pixel perfect match.

Components configurations

It will be required to provide extra configuration along with PSD document, similar to designerMediaConfig.json in Muse package, to make Ad unit interactive:

- for basic components: cta url, tracking id
    - x pinning
    - jump to second tab
    - mobile expandable features:
- for video: list of video/poster files
- for tabs component: text, trackers, colors, etc.
- for complex components: full configurations required, but we can use slides content for galleries

Layer’s border on particular side

We have a valid use case for this: Ad unit header section, where bottom border is configurable.

As a workaround we can create a group with rectangle shapes inside and predefined names:

"background" to get fill color, "border-bottom" to get bottom stroke only.

POC and results

Attached Vue component was used to generate header and tabs sections JSONs:





Default Advanced Campaign was created and preview mode opened in client showcase.

By use of Chrome’s dev tools response overrides feature, content were replaced with generated JSON.

Tab component configuration updated to reflect 2 tabs.

Mobile expandable Ad size was used, as it is the most complex one.

Result:


Dimensions: iPhone XR
414
x
896
100%
No throttling
vdx.tv
Captivate
viewers
Cop
y 1
vdx.tv
Lorem ipsum
dolor sit amet
Cop
y 1

Official APIs POC

There is possibility to execute raw commands by batchPlay API method:

https://developer.adobe.com/photoshop/uxp/2022/ps_reference/media/batchplay/

It's possible to export layer as png, jpg, etc. example code: https://github.com/AdobeDocs/uxp-photoshop-plugin-samples/blob/main/jszip-sample/src/index.js

And it's possible to fetch detailed information in JSON format be executing "get" command.

Also there is support of network communications fetch, WebSockets.

For plugin's UI elements it's possible to use UI libs Vue, React.

By use of simplified DOM API we were managed to patch existing project to reflect psd design:

- save layer's pixel data to png files into local filesystem
- upload this files as resources to predefined project
- patch project's items by matching psd layer.name === project's object.id

The code and plugin package:





Results PSD to VDX Studio



There is no extra information for partial text styling, but we can fetch this by use of raw "get" command.

What can we extract by raw "get" command?

Artboard JSON


```
[
  {
    "name": "inframe-970x250",
    "color": {
      "_enum": "color",
      "_value": "none"
    },
    "visible": true,
    "mode": {
      "_enum": "blendMode",
      "_value": "normal"
    },
    "opacity": 255,
    "layerID": 32,
    "itemIndex": 10,
    "count": 10,
    "preserveTransparency": false,
    "layerFXVisible": false,
    "globalAngle": 90,
    "background": false,
    "layerSection": {
      "_enum": "layerSectionType",
      "_value": "layerSectionStart"
```

```
},
"layerLocking": {
  "_obj": "layerLocking",
  "protectTransparency": false,
  "protectComposite": false,
  "protectPosition": false,
  "protectArtboardAutonest": false,
  "protectAll": false
},
"group": false,
"layerSectionExpanded": true,
"targetChannels": [
  {
    "_ref": "channel",
    "_index": 1
  },
  {
    "_ref": "channel",
    "_index": 2
  },
  {
    "_ref": "channel",
    "_index": 3
  }
],
"visibleChannels": [
  {
    "_ref": "channel",
    "_index": 1
  },
  {
    "_ref": "channel",
    "_index": 2
  },
  {
    "_ref": "channel",
    "_index": 3
  }
],
"channelRestrictions": [
  {
    "_enum": "channel",
    "_value": "red"
  },
  {
    "_enum": "channel",
    "_value": "grain"
  },
  {
    "_enum": "channel",
    "_value": "blue"
  }
],
"fillOpacity": 255,
"hasUserMask": false,
"hasVectorMask": false,
"proportionalScaling": false,
"layerKind": 7,
"hasFilterMask": false,
"userMaskDensity": 255,
"userMaskFeather": 0,
"vectorMaskDensity": 255,
"vectorMaskFeather": 0,
"filterMaskDensity": 255,
"filterMaskFeather": 0,
"bounds": {
  "_obj": "rectangle",
  "top": {
    "_unit": "pixelsUnit",
    "_value": 0
  },
```

```
    "left": {
      "_unit": "pixelsUnit",
      "_value": 0
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 250
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 970
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 970
    },
    "height": {
      "_unit": "pixelsUnit",
      "_value": 250
    }
  },
  "boundsNoEffects": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 0
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 0
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 250
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 970
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 970
    },
    "height": {
      "_unit": "pixelsUnit",
      "_value": 250
    }
  },
  "boundsNoMask": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 0
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 0
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 250
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 970
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 970
    },
    "height": {
```

```
        "_unit": "pixelsUnit",
        "_value": 250
      }
    },
    "useAlignedRendering": false,
    "generatorSettings": {
      "_obj": "generatorSettings"
    },
    "keyOriginType": [],
    "fillEnabled": false,
    "animationProtection": {
      "_obj": "animationProtection",
      "animationUnifyPosition": false,
      "animationUnifyEffects": false,
      "animationUnifyVisibility": false,
      "animationPropagate": true
    },
    "isSyntheticFill": false,
    "artboard": {
      "_obj": "artboard",
      "artboardRect": {
        "_obj": "classFloatRect",
        "top": 0,
        "left": 0,
        "bottom": 250,
        "right": 970
      },
      "guideIDs": [],
      "artboardPresetName": "",
      "color": {
        "_obj": "RGBColor",
        "red": 255,
        "grain": 255,
        "blue": 255
      },
      "artboardBackgroundType": 1
    },
    "artboardEnabled": true,
    "vectorMaskEmpty": true,
    "textWarningLevel": 0,
    "parentLayerID": -1
  }
]
```


Group JSON


```
[
  {
    "name": "cta",
    "color": {
      "_enum": "color",
      "_value": "none"
    },
    "visible": true,
    "mode": {
      "_enum": "blendMode",
      "_value": "passThrough"
    },
    "opacity": 255,
    "layerID": 38,
    "itemIndex": 7,
    "count": 10,
    "preserveTransparency": false,
    "layerFXVisible": false,
    "globalAngle": 90,
    "background": false,
    "layerSection": {
```

```
    "_enum": "layerSectionType",
    "_value": "layerSectionStart"
  },
  "layerLocking": {
    "_obj": "layerLocking",
    "protectTransparency": false,
    "protectComposite": false,
    "protectPosition": false,
    "protectArtboardAutonest": false,
    "protectAll": false
  },
  "group": false,
  "layerSectionExpanded": true,
  "targetChannels": [
    {
      "_ref": "channel",
      "_index": 1
    },
    {
      "_ref": "channel",
      "_index": 2
    },
    {
      "_ref": "channel",
      "_index": 3
    }
  ],
  "visibleChannels": [
    {
      "_ref": "channel",
      "_index": 1
    },
    {
      "_ref": "channel",
      "_index": 2
    },
    {
      "_ref": "channel",
      "_index": 3
    }
  ],
  "channelRestrictions": [
    {
      "_enum": "channel",
      "_value": "red"
    },
    {
      "_enum": "channel",
      "_value": "grain"
    },
    {
      "_enum": "channel",
      "_value": "blue"
    }
  ],
  "fillOpacity": 255,
  "hasUserMask": false,
  "hasVectorMask": false,
  "proportionalScaling": false,
  "layerKind": 7,
  "hasFilterMask": false,
  "userMaskDensity": 255,
  "userMaskFeather": 0,
  "vectorMaskDensity": 255,
  "vectorMaskFeather": 0,
  "filterMaskDensity": 255,
  "filterMaskFeather": 0,
  "bounds": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
```

```
      "_value": 0
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 0
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 250
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 970
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 970
    },
    "height": {
      "_unit": "pixelsUnit",
      "_value": 250
    }
  },
  "boundsNoEffects": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 192
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 516
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 226
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 644
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 128
    },
    "height": {
      "_unit": "pixelsUnit",
      "_value": 34
    }
  },
  "boundsNoMask": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 0
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 0
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 250
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 970
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 970
```

```
      },
      "height": {
        "_unit": "pixelsUnit",
        "_value": 250
      }
    },
    "useAlignedRendering": false,
    "generatorSettings": {
      "_obj": "generatorSettings"
    },
    "keyOriginType": [],
    "fillEnabled": false,
    "animationProtection": {
      "_obj": "animationProtection",
      "animationUnifyPosition": false,
      "animationUnifyEffects": false,
      "animationUnifyVisibility": false,
      "animationPropagate": true
    },
    "isSyntheticFill": false,
    "artboardEnabled": false,
    "vectorMaskEmpty": true,
    "textWarningLevel": 0,
    "parentLayerID": 32
  }
]
```


Layer JSON


```
[
  {
    "name": "logo",
    "color": {
      "_enum": "color",
      "_value": "none"
    },
    "visible": true,
    "mode": {
      "_enum": "blendMode",
      "_value": "normal"
    },
    "opacity": 255,
    "layerID": 8,
    "itemIndex": 3,
    "count": 10,
    "preserveTransparency": false,
    "layerFXVisible": false,
    "globalAngle": 90,
    "background": false,
    "layerSection": {
      "_enum": "layerSectionType",
      "_value": "layerSectionContent"
    },
    "layerLocking": {
      "_obj": "layerLocking",
      "protectTransparency": false,
      "protectComposite": false,
      "protectPosition": false,
      "protectArtboardAutonest": false,
      "protectAll": false
    },
    "group": false,
    "targetChannels": [
      {
        "_ref": "channel",
        "_index": 1
      },
```

```
    {
      "_ref": "channel",
      "_index": 2
    },
    {
      "_ref": "channel",
      "_index": 3
    }
  ],
  "visibleChannels": [
    {
      "_ref": "channel",
      "_index": 1
    },
    {
      "_ref": "channel",
      "_index": 2
    },
    {
      "_ref": "channel",
      "_index": 3
    }
  ],
  "channelRestrictions": [
    {
      "_enum": "channel",
      "_value": "red"
    },
    {
      "_enum": "channel",
      "_value": "grain"
    },
    {
      "_enum": "channel",
      "_value": "blue"
    }
  ],
  "fillOpacity": 255,
  "hasUserMask": false,
  "hasVectorMask": false,
  "proportionalScaling": false,
  "layerKind": 1,
  "hasFilterMask": false,
  "userMaskDensity": 255,
  "userMaskFeather": 0,
  "vectorMaskDensity": 255,
  "vectorMaskFeather": 0,
  "filterMaskDensity": 255,
  "filterMaskFeather": 0,
  "bounds": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 82
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 643
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 123
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 778
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 135
    },
```

```
    "height": {
      "_unit": "pixelsUnit",
      "_value": 41
    }
  },
  "boundsNoEffects": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 82
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 643
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 123
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 778
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 135
    },
    "height": {
      "_unit": "pixelsUnit",
      "_value": 41
    }
  },
  "boundsNoMask": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 82
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 643
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 123
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 778
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 135
    },
    "height": {
      "_unit": "pixelsUnit",
      "_value": 41
    }
  },
  "useAlignedRendering": false,
  "generatorSettings": {
    "_obj": "generatorSettings"
  },
  "keyOriginType": [],
  "fillEnabled": false,
  "animationProtection": {
    "_obj": "animationProtection",
    "animationUnifyPosition": false,
    "animationUnifyEffects": false,
    "animationUnifyVisibility": false,
    "animationPropagate": true
```

```
    },
    "isSyntheticFill": false,
    "artboardEnabled": false,
    "vectorMaskEmpty": true,
    "textWarningLevel": 0,
    "parentLayerID": 32
  }
]
```


Text Layer JSON


```
[
  {
    "name": "copy2",
    "color": {
      "_enum": "color",
      "_value": "none"
    },
    "visible": true,
    "mode": {
      "_enum": "blendMode",
      "_value": "normal"
    },
    "opacity": 255,
    "layerID": 13,
    "itemIndex": 8,
    "count": 10,
    "preserveTransparency": false,
    "layerFXVisible": false,
    "globalAngle": 90,
    "background": false,
    "textKey": {
      "_obj": "textLayer",
      "textKey": "customer-centric company",
      "warp": {
        "_obj": "warp",
        "warpStyle": {
          "_enum": "warpStyle",
          "_value": "warpNone"
        },
        "warpValue": 0,
        "warpPerspective": 0,
        "warpPerspectiveOther": 0,
        "warpRotate": {
          "_enum": "orientation",
          "_value": "horizontal"
        }
      },
      "textClickPoint": {
        "_obj": "paint",
        "horizontal": {
          "_unit": "percentUnit",
          "_value": 47.10615335051546
        },
        "vertical": {
          "_unit": "percentUnit",
          "_value": 56.1046875
        }
      },
      "textGridding": {
        "_enum": "textGridding",
        "_value": "none"
      },
      "orientation": {
        "_enum": "orientation",
        "_value": "horizontal"
      },
```

```
"antiAlias": {
  "_enum": "antiAliasType",
  "_value": "antiAliasSmooth"
},
"bounds": {
  "_obj": "bounds",
  "left": {
    "_unit": "pixelsUnit",
    "_value": 0
  },
  "top": {
    "_unit": "pixelsUnit",
    "_value": -20.595703125
  },
  "right": {
    "_unit": "pixelsUnit",
    "_value": 281.42578125
  },
  "bottom": {
    "_unit": "pixelsUnit",
    "_value": 7.79296875
  }
},
"boundingBox": {
  "_obj": "boundingBox",
  "left": {
    "_unit": "pixelsUnit",
    "_value": 1
  },
  "top": {
    "_unit": "pixelsUnit",
    "_value": -17.34375
  },
  "right": {
    "_unit": "pixelsUnit",
    "_value": 280.50390625
  },
  "bottom": {
    "_unit": "pixelsUnit",
    "_value": 5
  }
},
"textShape": [
  {
    "_obj": "textShape",
    "char": {
      "_enum": "char",
      "_value": "paint"
    },
    "orientation": {
      "_enum": "orientation",
      "_value": "horizontal"
    },
    "transform": {
      "_obj": "transform",
      "xx": 1,
      "xy": 0,
      "yx": 0,
      "yy": 1,
      "tx": 0,
      "ty": 0
    },
    "rowCount": 1,
    "columnCount": 1,
    "rowMajorOrder": true,
    "rowGutter": {
      "_unit": "pixelsUnit",
      "_value": 0
    },
    "columnGutter": {
      "_unit": "pixelsUnit",
```

```
        "_value": 0
      },
      "spacing": {
        "_unit": "pixelsUnit",
        "_value": 0
      },
      "frameBaselineAlignment": {
        "_enum": "frameBaselineAlignment",
        "_value": "alignByAscent"
      },
      "firstBaselineMinimum": {
        "_unit": "pixelsUnit",
        "_value": 0
      },
      "base": {
        "_obj": "paint",
        "horizontal": 0,
        "vertical": 0
      }
    }
  ],
  "textStyleRange": [
    {
      "_obj": "textStyleRange",
      "from": 0,
      "to": 5,
      "textStyle": {
        "_obj": "textStyle",
        "styleSheetHasParent": true,
        "fontPostScriptName": "ArialMT",
        "fontName": "Arial",
        "fontStyleName": "Regular",
        "fontScript": 0,
        "fontTechnology": 1,
        "fontAvailable": true,
        "size": {
          "_unit": "pixelsUnit",
          "_value": 24
        },
        "impliedFontSize": {
          "_unit": "pixelsUnit",
          "_value": 24
        },
        "horizontalScale": 100,
        "verticalScale": 100,
        "syntheticBold": false,
        "syntheticItalic": false,
        "autoLeading": false,
        "leading": {
          "_unit": "pixelsUnit",
          "_value": 16
        },
        "impliedLeading": {
          "_unit": "pixelsUnit",
          "_value": 16
        },
        "tracking": 0,
        "baselineShift": {
          "_unit": "pixelsUnit",
          "_value": 0
        },
        "impliedBaselineShift": {
          "_unit": "pixelsUnit",
          "_value": 0
        },
        "autoKern": {
          "_enum": "autoKern",
          "_value": "manual"
        },
        "fontCaps": {
          "_enum": "fontCaps",
```

```
    "_value": "normal"
  },
  "digitSet": {
    "_enum": "digitSet",
    "_value": "defaultDigits"
  },
  "kashidas": {
    "_enum": "kashidas",
    "_value": "kashidaDefault"
  },
  "diacXOffset": {
    "_unit": "pixelsUnit",
    "_value": 0
  },
  "diacYOffset": {
    "_unit": "pixelsUnit",
    "_value": 0
  },
  "markYDistFromBaseline": {
    "_unit": "pixelsUnit",
    "_value": 0.0005499999970197678
  },
  "baseline": {
    "_enum": "baseline",
    "_value": "normal"
  },
  "otbaseline": {
    "_enum": "otbaseline",
    "_value": "normal"
  },
  "strikethrough": {
    "_enum": "strikethrough",
    "_value": "strikethroughOff"
  },
  "underline": {
    "_enum": "underline",
    "_value": "underlineOff"
  },
  "ligature": true,
  "altligature": false,
  "contextualLigatures": true,
  "fractions": false,
  "ordinals": false,
  "swash": false,
  "titling": false,
  "connectionForms": false,
  "stylisticAlternates": false,
  "stylisticSets": 0,
  "ornaments": false,
  "justificationAlternates": false,
  "figureStyle": {
    "_enum": "figureStyle",
    "_value": "normal"
  },
  "proportionalMetrics": false,
  "kana": false,
  "italics": false,
  "baselineDirection": {
    "_enum": "baselineDirection",
    "_value": "withStream"
  },
  "textLanguage": {
    "_enum": "textLanguage",
    "_value": "englishLanguage"
  },
  "japaneseAlternate": {
    "_enum": "japaneseAlternate",
    "_value": "defaultForm"
  },
  "mojiZume": 0,
  "gridAlignment": {
```

```
    "_enum": "gridAlignment",
    "_value": "roman"
  },
  "noBreak": false,
  "color": {
    "_obj": "RGBColor",
    "red": 247.0006388425827,
    "grain": 147.99689054489136,
    "blue": 0
  },
  "strokeColor": {
    "_obj": "RGBColor",
    "red": 0,
    "grain": 0,
    "blue": 0
  },
  "baseParentStyle": {
    "_obj": "textStyle",
    "fontPostScriptName": "MyriadPro-Regular",
    "fontName": "Myriad Pro",
    "fontStyleName": "Regular",
    "fontScript": 0,
    "fontTechnology": 0,
    "fontAvailable": true,
    "size": {
      "_unit": "pixelsUnit",
      "_value": 12
    },
    "impliedFontSize": {
      "_unit": "pixelsUnit",
      "_value": 12
    },
    "horizontalScale": 100,
    "verticalScale": 100,
    "syntheticBold": false,
    "syntheticItalic": false,
    "autoLeading": true,
    "tracking": 0,
    "baselineShift": {
      "_unit": "pixelsUnit",
      "_value": 0
    },
    "impliedBaselineShift": {
      "_unit": "pixelsUnit",
      "_value": 0
    },
    "characterRotation": 0,
    "autoKern": {
      "_enum": "autoKern",
      "_value": "metricsKern"
    },
    "fontCaps": {
      "_enum": "fontCaps",
      "_value": "normal"
    },
    "digitSet": {
      "_enum": "digitSet",
      "_value": "defaultDigits"
    },
    "dirOverride": {
      "_enum": "dirOverride",
      "_value": "dirOverrideDefault"
    },
    "kashidas": {
      "_enum": "kashidas",
      "_value": "kashidaDefault"
    },
    "diacVPos": {
      "_enum": "diacVPos",
      "_value": "diacVPosOpenType"
    },
```

```
"diacXOffset": {
  "_unit": "pixelsUnit",
  "_value": 0
},
"diacYOffset": {
  "_unit": "pixelsUnit",
  "_value": 0
},
"markYDistFromBaseline": {
  "_unit": "pixelsUnit",
  "_value": 100
},
"baseline": {
  "_enum": "baseline",
  "_value": "normal"
},
"otbaseline": {
  "_enum": "otbaseline",
  "_value": "normal"
},
"strikethrough": {
  "_enum": "strikethrough",
  "_value": "strikethroughOff"
},
"underline": {
  "_enum": "underline",
  "_value": "underlineOff"
},
"underlineOffset": {
  "_unit": "pixelsUnit",
  "_value": 0
},
"ligature": true,
"altligature": false,
"contextualLigatures": false,
"alternateLigatures": false,
"oldStyle": false,
"fractions": false,
"ordinals": false,
"swash": false,
"titling": false,
"connectionForms": false,
"stylisticAlternates": false,
"stylisticSets": 0,
"ornaments": false,
"justificationAlternates": false,
"figureStyle": {
  "_enum": "figureStyle",
  "_value": "normal"
},
"proportionalMetrics": false,
"kana": false,
"italics": false,
"ruby": false,
"baselineDirection": {
  "_enum": "baselineDirection",
  "_value": "rotated"
},
"textLanguage": {
  "_enum": "textLanguage",
  "_value": "englishLanguage"
},
"japaneseAlternate": {
  "_enum": "japaneseAlternate",
  "_value": "defaultForm"
},
"mojiZume": 0,
"gridAlignment": {
  "_enum": "gridAlignment",
  "_value": "roman"
},
```

```
        "enableWariChu": false,
        "wariChuCount": 2,
        "wariChuLineGap": 0,
        "wariChuScale": 0.5,
        "wariChuWidow": 2,
        "wariChuOrphan": 2,
        "wariChuJustification": {
          "_enum": "wariChuJustification",
          "_value": "wariChuAutoJustify"
        },
        "tcyUpDown": 0,
        "tcyLeftRight": 0,
        "leftAki": -1,
        "rightAki": -1,
        "jiDori": 0,
        "noBreak": false,
        "color": {
          "_obj": "RGBColor",
          "red": 0,
          "grain": 0,
          "blue": 0
        },
        "strokeColor": {
          "_obj": "RGBColor",
          "red": 0,
          "grain": 0,
          "blue": 0
        },
        "fill": true,
        "stroke": false,
        "fillFirst": true,
        "fillOverPrint": false,
        "strokeOverPrint": false,
        "lineCap": {
          "_enum": "lineCap",
          "_value": "buttCap"
        },
        "lineJoin": {
          "_enum": "lineJoin",
          "_value": "miterJoin"
        },
        "lineWidth": {
          "_unit": "pixelsUnit",
          "_value": 1
        },
        "miterLimit": {
          "_unit": "pixelsUnit",
          "_value": 4
        },
        "lineDashoffset": 0
      }
    }
  },
  {
    "_obj": "textStyleRange",
    "from": 5,
    "to": 6,
    "textStyle": {
      "_obj": "textStyle",
      "styleSheetHasParent": true,
      "fontPostScriptName": "ArialMT",
      "fontName": "Arial",
      "fontStyleName": "Regular",
      "fontScript": 0,
      "fontTechnology": 1,
      "fontAvailable": true,
      "size": {
        "_unit": "pixelsUnit",
        "_value": 24
      },
      "impliedFontSize": {
```

```
    "_unit": "pixelsUnit",
    "_value": 24
  },
  "horizontalScale": 100,
  "verticalScale": 100,
  "syntheticBold": false,
  "syntheticItalic": false,
  "autoLeading": false,
  "leading": {
    "_unit": "pixelsUnit",
    "_value": 16
  },
  "impliedLeading": {
    "_unit": "pixelsUnit",
    "_value": 16
  },
  "tracking": 0,
  "baselineShift": {
    "_unit": "pixelsUnit",
    "_value": 0
  },
  "impliedBaselineShift": {
    "_unit": "pixelsUnit",
    "_value": 0
  },
  "autoKern": {
    "_enum": "autoKern",
    "_value": "manual"
  },
  "fontCaps": {
    "_enum": "fontCaps",
    "_value": "normal"
  },
  "digitSet": {
    "_enum": "digitSet",
    "_value": "defaultDigits"
  },
  "kashidas": {
    "_enum": "kashidas",
    "_value": "kashidaDefault"
  },
  "diacXOffset": {
    "_unit": "pixelsUnit",
    "_value": 0
  },
  "diacYOffset": {
    "_unit": "pixelsUnit",
    "_value": 0
  },
  "markYDistFromBaseline": {
    "_unit": "pixelsUnit",
    "_value": 0.0005499999970197678
  },
  "baseline": {
    "_enum": "baseline",
    "_value": "normal"
  },
  "otbaseline": {
    "_enum": "otbaseline",
    "_value": "normal"
  },
  "strikethrough": {
    "_enum": "strikethrough",
    "_value": "strikethroughOff"
  },
  "underline": {
    "_enum": "underline",
    "_value": "underlineOff"
  },
  "ligature": true,
  "altligature": false,
```

```
      "contextualLigatures": true,
      "fractions": false,
      "ordinals": false,
      "swash": false,
      "titling": false,
      "connectionForms": false,
      "stylisticAlternates": false,
      "stylisticSets": 0,
      "ornaments": false,
      "justificationAlternates": false,
      "figureStyle": {
        "_enum": "figureStyle",
        "_value": "normal"
      },
      "proportionalMetrics": false,
      "kana": false,
      "italics": false,
      "baselineDirection": {
        "_enum": "baselineDirection",
        "_value": "withStream"
      },
      "textLanguage": {
        "_enum": "textLanguage",
        "_value": "englishLanguage"
      },
      "japaneseAlternate": {
        "_enum": "japaneseAlternate",
        "_value": "defaultForm"
      },
      "mojiZume": 0,
      "gridAlignment": {
        "_enum": "gridAlignment",
        "_value": "roman"
      },
      "noBreak": false,
      "color": {
        "_obj": "RGBColor",
        "red": 26.800501495599747,
        "grain": 71.89979508519173,
        "blue": 201.0011887550354
      },
      "strokeColor": {
        "_obj": "RGBColor",
        "red": 0,
        "grain": 0,
        "blue": 0
      }
    }
  },
  {
    "_obj": "textStyleRange",
    "from": 6,
    "to": 25,
    "textStyle": {
      "_obj": "textStyle",
      "styleSheetHasParent": true,
      "fontPostScriptName": "ArialMT",
      "fontName": "Arial",
      "fontStyleName": "Regular",
      "fontScript": 0,
      "fontTechnology": 1,
      "fontAvailable": true,
      "size": {
        "_unit": "pixelsUnit",
        "_value": 24
      },
      "impliedFontSize": {
        "_unit": "pixelsUnit",
        "_value": 24
      },
      "horizontalScale": 100,
```

```
"verticalScale": 100,
"syntheticBold": false,
"syntheticItalic": false,
"autoLeading": false,
"leading": {
  "_unit": "pixelsUnit",
  "_value": 16
},
"impliedLeading": {
  "_unit": "pixelsUnit",
  "_value": 16
},
"tracking": 0,
"baselineShift": {
  "_unit": "pixelsUnit",
  "_value": 0
},
"impliedBaselineShift": {
  "_unit": "pixelsUnit",
  "_value": 0
},
"autoKern": {
  "_enum": "autoKern",
  "_value": "manual"
},
"fontCaps": {
  "_enum": "fontCaps",
  "_value": "normal"
},
"digitSet": {
  "_enum": "digitSet",
  "_value": "defaultDigits"
},
"kashidas": {
  "_enum": "kashidas",
  "_value": "kashidaDefault"
},
"diacXOffset": {
  "_unit": "pixelsUnit",
  "_value": 0
},
"diacYOffset": {
  "_unit": "pixelsUnit",
  "_value": 0
},
"markYDistFromBaseline": {
  "_unit": "pixelsUnit",
  "_value": 0.0005499999970197678
},
"baseline": {
  "_enum": "baseline",
  "_value": "normal"
},
"otbaseline": {
  "_enum": "otbaseline",
  "_value": "normal"
},
"strikethrough": {
  "_enum": "strikethrough",
  "_value": "strikethroughOff"
},
"underline": {
  "_enum": "underline",
  "_value": "underlineOff"
},
"ligature": true,
"altligature": false,
"contextualLigatures": true,
"fractions": false,
"ordinals": false,
"swash": false,
```

```
        "titling": false,
        "connectionForms": false,
        "stylisticAlternates": false,
        "stylisticSets": 0,
        "ornaments": false,
        "justificationAlternates": false,
        "figureStyle": {
          "_enum": "figureStyle",
          "_value": "normal"
        },
        "proportionalMetrics": false,
        "kana": false,
        "italics": false,
        "baselineDirection": {
          "_enum": "baselineDirection",
          "_value": "withStream"
        },
        "textLanguage": {
          "_enum": "textLanguage",
          "_value": "englishLanguage"
        },
        "japaneseAlternate": {
          "_enum": "japaneseAlternate",
          "_value": "defaultForm"
        },
        "mojiZume": 0,
        "gridAlignment": {
          "_enum": "gridAlignment",
          "_value": "roman"
        },
        "noBreak": false,
        "color": {
          "_obj": "RGBColor",
          "red": 247.0006388425827,
          "grain": 147.99689054489136,
          "blue": 0
        },
        "strokeColor": {
          "_obj": "RGBColor",
          "red": 0,
          "grain": 0,
          "blue": 0
        }
      }
    }
  ],
  "paragraphStyleRange": [
    {
      "_obj": "paragraphStyleRange",
      "from": 0,
      "to": 25,
      "paragraphStyle": {
        "_obj": "paragraphStyle",
        "styleSheetHasParent": true,
        "align": {
          "_enum": "alignmentType",
          "_value": "left"
        },
        "directionType": {
          "_enum": "directionType",
          "_value": "dirLeftToRight"
        },
        "textComposerEngine": {
          "_enum": "textComposerEngine",
          "_value": "textOptycaComposer"
        },
        "defaultStyle": {
          "_obj": "textStyle",
          "fontPostScriptName": "MyriadPro-Regular",
          "fontName": "Myriad Pro",
          "fontStyleName": "Regular",
```

```
"fontScript": 0,
"fontTechnology": 0,
"fontAvailable": true,
"size": {
  "_unit": "pixelsUnit",
  "_value": 12
},
"horizontalScale": 100,
"verticalScale": 100,
"syntheticBold": false,
"syntheticItalic": false,
"autoLeading": true,
"tracking": 0,
"baselineShift": {
  "_unit": "pixelsUnit",
  "_value": 0
},
"characterRotation": 0,
"autoKern": {
  "_enum": "autoKern",
  "_value": "metricsKern"
},
"fontCaps": {
  "_enum": "fontCaps",
  "_value": "normal"
},
"digitSet": {
  "_enum": "digitSet",
  "_value": "arabicDigits"
},
"kashidas": {
  "_enum": "kashidas",
  "_value": "kashidaDefault"
},
"diacVPos": {
  "_enum": "diacVPos",
  "_value": "diacVPosOpenType"
},
"diacXOffset": {
  "_unit": "pixelsUnit",
  "_value": 0
},
"diacYOffset": {
  "_unit": "pixelsUnit",
  "_value": 0
},
"markYDistFromBaseline": {
  "_unit": "pixelsUnit",
  "_value": 0
},
"baseline": {
  "_enum": "baseline",
  "_value": "normal"
},
"strikethrough": {
  "_enum": "strikethrough",
  "_value": "strikethroughOff"
},
"underline": {
  "_enum": "underline",
  "_value": "underlineOff"
},
"ligature": true,
"altligature": false,
"contextualLigatures": true,
"alternateLigatures": false,
"oldStyle": false,
"fractions": false,
"ordinals": false,
"swash": false,
"titling": false,
```

```
            "connectionForms": true,
            "stylisticAlternates": false,
            "stylisticSets": 0,
            "ornaments": false,
            "figureStyle": {
              "_enum": "figureStyle",
              "_value": "normal"
            },
            "baselineDirection": {
              "_enum": "baselineDirection",
              "_value": "withStream"
            },
            "textLanguage": {
              "_enum": "textLanguage",
              "_value": "englishLanguage"
            },
            "color": {
              "_obj": "RGBColor",
              "red": 0,
              "grain": 0,
              "blue": 0
            },
            "strokeColor": {
              "_obj": "RGBColor",
              "red": 0,
              "grain": 0,
              "blue": 0
            }
          }
        }
      }
    ],
    "kerningRange": []
  },
  "layerSection": {
    "_enum": "layerSectionType",
    "_value": "layerSectionContent"
  },
  "layerLocking": {
    "_obj": "layerLocking",
    "protectTransparency": false,
    "protectComposite": false,
    "protectPosition": false,
    "protectArtboardAutonest": false,
    "protectAll": false
  },
  "group": false,
  "targetChannels": [
    {
      "_ref": "channel",
      "_index": 1
    },
    {
      "_ref": "channel",
      "_index": 2
    },
    {
      "_ref": "channel",
      "_index": 3
    }
  ],
  "visibleChannels": [
    {
      "_ref": "channel",
      "_index": 1
    },
    {
      "_ref": "channel",
      "_index": 2
    },
    {
```

```
      "_ref": "channel",
      "_index": 3
    }
  ],
  "channelRestrictions": [
    {
      "_enum": "channel",
      "_value": "red"
    },
    {
      "_enum": "channel",
      "_value": "grain"
    },
    {
      "_enum": "channel",
      "_value": "blue"
    }
  ],
  "fillOpacity": 255,
  "hasUserMask": false,
  "hasVectorMask": false,
  "proportionalScaling": false,
  "layerKind": 3,
  "hasFilterMask": false,
  "userMaskDensity": 255,
  "userMaskFeather": 0,
  "vectorMaskDensity": 255,
  "vectorMaskFeather": 0,
  "filterMaskDensity": 255,
  "filterMaskFeather": 0,
  "bounds": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 123
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 458
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 146
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 738
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 280
    },
    "height": {
      "_unit": "pixelsUnit",
      "_value": 23
    }
  },
  "boundsNoEffects": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 123
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 458
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 146
    },
```

```
    "right": {
      "_unit": "pixelsUnit",
      "_value": 738
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 280
    },
    "height": {
      "_unit": "pixelsUnit",
      "_value": 23
    }
  },
  "boundsNoMask": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 123
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 458
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 146
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 738
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 280
    },
    "height": {
      "_unit": "pixelsUnit",
      "_value": 23
    }
  },
  "boundingBox": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 123
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 458
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 145
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 737
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 279
    },
    "height": {
      "_unit": "pixelsUnit",
      "_value": 22
    }
  },
  "useAlignedRendering": false,
  "generatorSettings": {
    "_obj": "generatorSettings"
  },
```

```
    "keyOriginType": [],
    "fillEnabled": false,
    "animationProtection": {
      "_obj": "animationProtection",
      "animationUnifyPosition": false,
      "animationUnifyEffects": false,
      "animationUnifyVisibility": false,
      "animationPropagate": true
    },
    "isSyntheticFill": false,
    "artboardEnabled": false,
    "vectorMaskEmpty": true,
    "textWarningLevel": 0,
    "parentLayerID": 32
  }
]
```


Rectangle Layer JSON


```
[
  {
    "name": "background",
    "color": {
      "_enum": "color",
      "_value": "none"
    },
    "visible": true,
    "mode": {
      "_enum": "blendMode",
      "_value": "normal"
    },
    "opacity": 255,
    "layerID": 10,
    "itemIndex": 5,
    "count": 10,
    "preserveTransparency": false,
    "layerFXVisible": false,
    "globalAngle": 90,
    "background": false,
    "layerSection": {
      "_enum": "layerSectionType",
      "_value": "layerSectionContent"
    },
    "layerLocking": {
      "_obj": "layerLocking",
      "protectTransparency": false,
      "protectComposite": false,
      "protectPosition": false,
      "protectArtboardAutonest": false,
      "protectAll": false
    },
    "group": false,
    "targetChannels": [
      {
        "_ref": "channel",
        "_index": 1
      },
      {
        "_ref": "channel",
        "_index": 2
      },
      {
        "_ref": "channel",
        "_index": 3
      }
    ],
    "visibleChannels": [
```

```
    {
      "_ref": "channel",
      "_index": 1
    },
    {
      "_ref": "channel",
      "_index": 2
    },
    {
      "_ref": "channel",
      "_index": 3
    }
  ],
  "channelRestrictions": [
    {
      "_enum": "channel",
      "_value": "red"
    },
    {
      "_enum": "channel",
      "_value": "grain"
    },
    {
      "_enum": "channel",
      "_value": "blue"
    }
  ],
  "fillOpacity": 255,
  "hasUserMask": false,
  "hasVectorMask": true,
  "proportionalScaling": false,
  "layerKind": 4,
  "hasFilterMask": false,
  "userMaskDensity": 255,
  "userMaskFeather": 0,
  "vectorMaskDensity": 255,
  "vectorMaskFeather": 0,
  "filterMaskDensity": 255,
  "filterMaskFeather": 0,
  "adjustment": [
    {
      "_obj": "solidColorLayer",
      "color": {
        "_obj": "RGBColor",
        "red": 255,
        "grain": 153.00155639648438,
        "blue": 0
      }
    }
  ],
  "bounds": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 192
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 516
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 226
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 644
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 128
```

```
    },
    "height": {
      "_unit": "pixelsUnit",
      "_value": 34
    }
  },
  "boundsNoEffects": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 192
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 516
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 226
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 644
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 128
    },
    "height": {
      "_unit": "pixelsUnit",
      "_value": 34
    }
  },
  "boundsNoMask": {
    "_obj": "rectangle",
    "top": {
      "_unit": "pixelsUnit",
      "_value": 0
    },
    "left": {
      "_unit": "pixelsUnit",
      "_value": 0
    },
    "bottom": {
      "_unit": "pixelsUnit",
      "_value": 250
    },
    "right": {
      "_unit": "pixelsUnit",
      "_value": 970
    },
    "width": {
      "_unit": "pixelsUnit",
      "_value": 970
    },
    "height": {
      "_unit": "pixelsUnit",
      "_value": 250
    }
  },
  "pathBounds": {
    "_obj": "rectangle",
    "pathBounds": {
      "_obj": "classFloatRect",
      "top": 192.375,
      "left": 516.375,
      "bottom": 225.375,
      "right": 643.21875
    }
  },
  "useAlignedRendering": true,
```

```
"generatorSettings": {
  "_obj": "generatorSettings"
},
"AGMStrokeStyleInfo": {
  "_obj": "strokeStyle",
  "strokeStyleVersion": 2,
  "strokeEnabled": true,
  "fillEnabled": true,
  "strokeStyleLineWidth": {
    "_unit": "pixelsUnit",
    "_value": 1
  },
  "strokeStyleLineDashOffset": {
    "_unit": "pointsUnit",
    "_value": 0
  },
  "strokeStyleMiterLimit": 100,
  "strokeStyleLineCapType": {
    "_enum": "strokeStyleLineCapType",
    "_value": "strokeStyleButtCap"
  },
  "strokeStyleLineJoinType": {
    "_enum": "strokeStyleLineJoinType",
    "_value": "strokeStyleMiterJoin"
  },
  "strokeStyleLineAlignment": {
    "_enum": "strokeStyleLineAlignment",
    "_value": "strokeStyleAlignCenter"
  },
  "strokeStyleScaleLock": false,
  "strokeStyleStrokeAdjust": false,
  "strokeStyleLineDashSet": [],
  "strokeStyleBlendMode": {
    "_enum": "blendMode",
    "_value": "normal"
  },
  "strokeStyleOpacity": {
    "_unit": "percentUnit",
    "_value": 100
  },
  "strokeStyleContent": {
    "_obj": "solidColorLayer",
    "color": {
      "_obj": "RGBColor",
      "red": 255,
      "grain": 255,
      "blue": 255
    }
  },
  "strokeStyleResolution": 72
},
"keyOriginType": [
  {
    "keyOriginType": 2,
    "keyOriginRRectRadii": {
      "_obj": "radii",
      "unitValueQuadVersion": 1,
      "topRight": {
        "_unit": "pixelsUnit",
        "_value": 15
      },
      "topLeft": {
        "_unit": "pixelsUnit",
        "_value": 15
      },
      "bottomLeft": {
        "_unit": "pixelsUnit",
        "_value": 15
      },
      "bottomRight": {
        "_unit": "pixelsUnit",
```

```
        "_value": 15
      }
    },
    "keyOriginResolution": 72,
    "keyOriginShapeBBox": {
      "_obj": "unitRect",
      "unitValueQuadVersion": 1,
      "top": {
        "_unit": "pixelsUnit",
        "_value": 192.8637552460644
      },
      "left": {
        "_unit": "pixelsUnit",
        "_value": 516.8637552460643
      },
      "bottom": {
        "_unit": "pixelsUnit",
        "_value": 224.87421689972837
      },
      "right": {
        "_unit": "pixelsUnit",
        "_value": 642.723070384334
      }
    },
    "keyOriginBoxCorners": {
      "rectangleCornerA": {
        "_obj": "paint",
        "horizontal": 516.8637552460643,
        "vertical": 192.8637552460644
      },
      "rectangleCornerB": {
        "_obj": "paint",
        "horizontal": 642.723070384334,
        "vertical": 192.8637552460644
      },
      "rectangleCornerC": {
        "_obj": "paint",
        "horizontal": 642.723070384334,
        "vertical": 224.87421689972837
      },
      "rectangleCornerD": {
        "_obj": "paint",
        "horizontal": 516.8637552460643,
        "vertical": 224.87421689972837
      }
    },
    "transform": {
      "_obj": "transform",
      "xx": 0.7275104921287263,
      "xy": 0,
      "yx": 0,
      "yy": 0.7275104921287266,
      "tx": 166.93120853214697,
      "ty": 70.6419925684383
    },
    "keyOriginIndex": 0,
    "keyActionMode": 0
  }
],
"fillEnabled": true,
"animationProtection": {
  "_obj": "animationProtection",
  "animationUnifyPosition": false,
  "animationUnifyEffects": false,
  "animationUnifyVisibility": false,
  "animationPropagate": true
},
"isSyntheticFill": false,
"artboardEnabled": false,
"vectorMaskEnabled": true,
"vectorMaskEmpty": false,
```

```
    "vectorMaskLinked": true,
    "textWarningLevel": 0,
    "parentLayerID": 38
  }
]
```

Conclusion

I'm sure we need to go with official API. We can implement a "VDX Turbo" plugin for Photoshop, where user will be able to sign in, browse turbo's projects and do design synchronisations and much more.

We can use React for this plugin UI. Or we can still just compose a zip file with all the details in JSON and sliced assets, which can be later uploaded into Turbo.